quantms-rescoring and multi-engine searching enable deep proteome coverage across quantitative proteomics, immunopeptidomics, and phosphoproteomics

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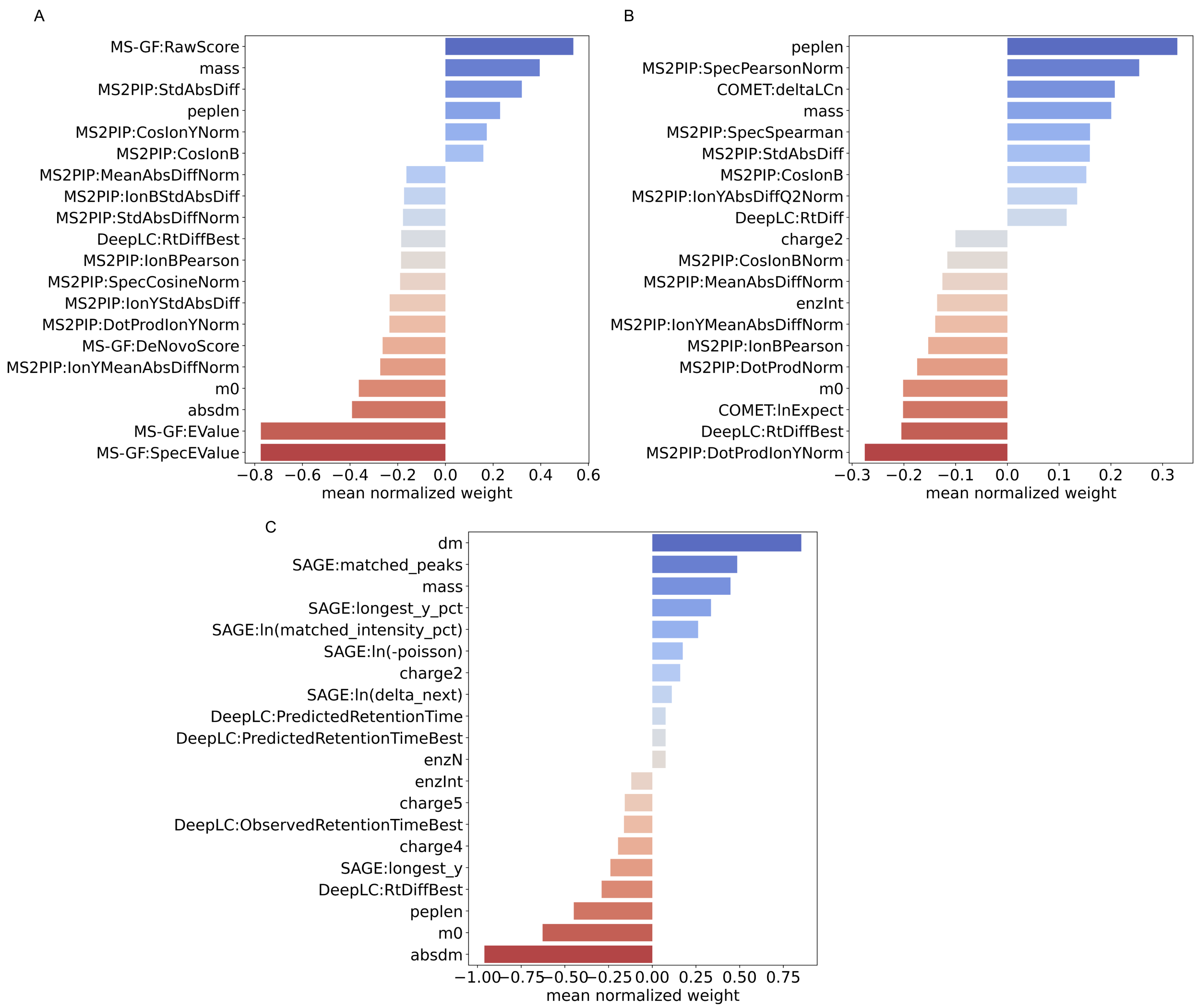

Figure S1: The weights that Percolator assigns to different features of the search engine feature vector in MSGF+ (A), Comet (B), and SAGE (C) combined with Percolator settings from PXD001819. The more different from zero a weight is (both positive and negative), the more important that feature is in the final Percolator classification.

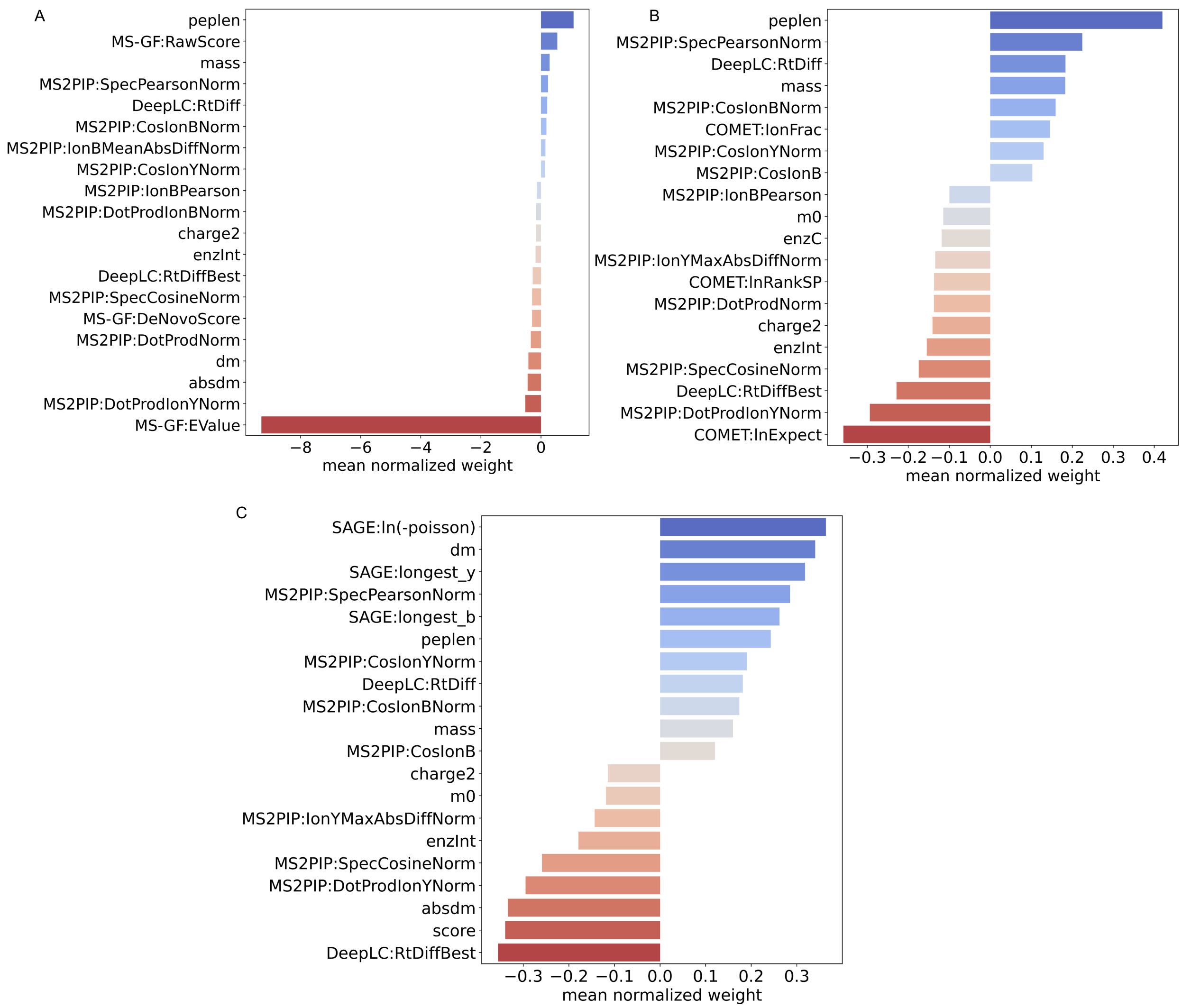

Figure S2: The top 20 normalized weights from Percolator for MSGF+ (A), Comet (B), and SAGE (C) identification results from PDC000125.

Table S1: Spectrum-quality features. The memory and CPU values represent the maximum resources required for a single job. The quantms workflow was run in a 64 cluster queue.

Features	**Definition**
signal to noise	The signal-to-noise ratio calculated as the maximum intensity divided by the root mean square deviation of the intensities for a spectrum.
spectral entropy	The entropy of normalized intensity
fraction tic top 10	The intensity fraction of top 10 peaks for total ion current
weighted std mz	The standard deviation of weighted m/z

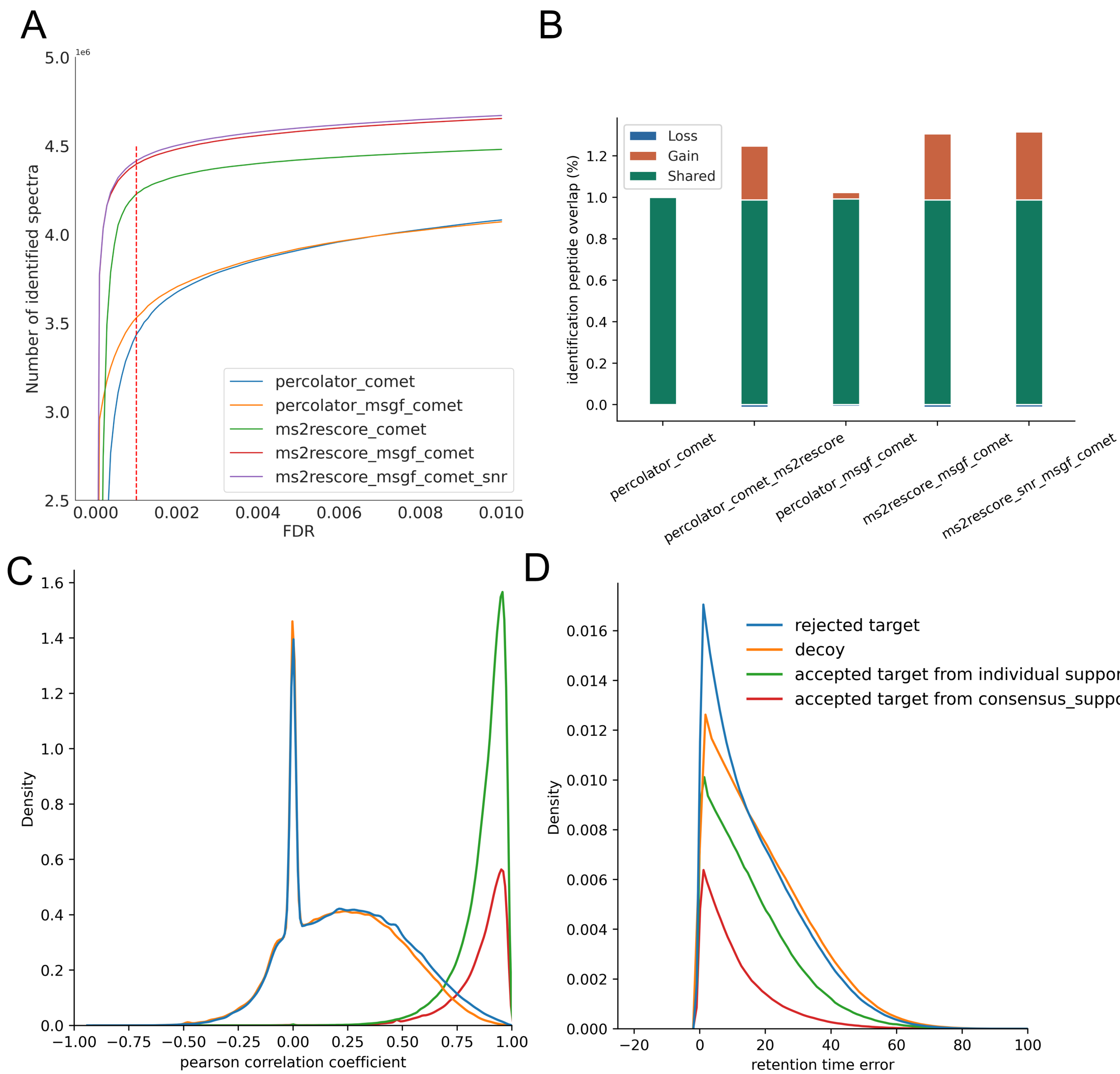

Figure S3: Comparison of immunopeptides identification results for different workflow settings on PXD019643. (A) Line plot showing the number of spectra identified for four workflow settings at different PSM FDR levels. (B) Percentage of unique identified peptides using different workflow settings. Density plots showing the distribution of the smallest retention time error between observed and predicted retention time (C) and the Pearson correlation between observed and predicted peak intensities (D) for each PSM split into decoys (red), rejected targets with q-value >0.01 (blue), and accepted targets from individual support and consensus support with q-value <0.01 (green). Note that the rejected target distribution coincides with the decoy distribution. The pronounced spike at Pearson correlation = 0 reflects spectrum comparisons in which all predicted or observed peaks have zero intensity, causing a division-by-zero situation that leaves the Pearson correlation undefined.

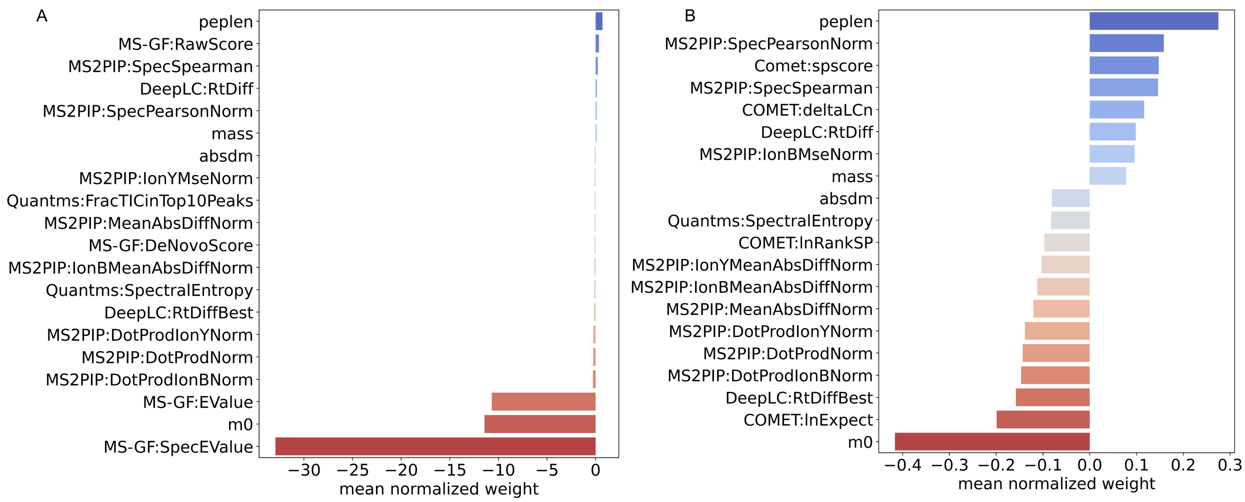

Figure S4: The top 20 normalized weights from Percolator for MSGF+ (A) and Comet (B) identification results in the PXD019643 dataset.

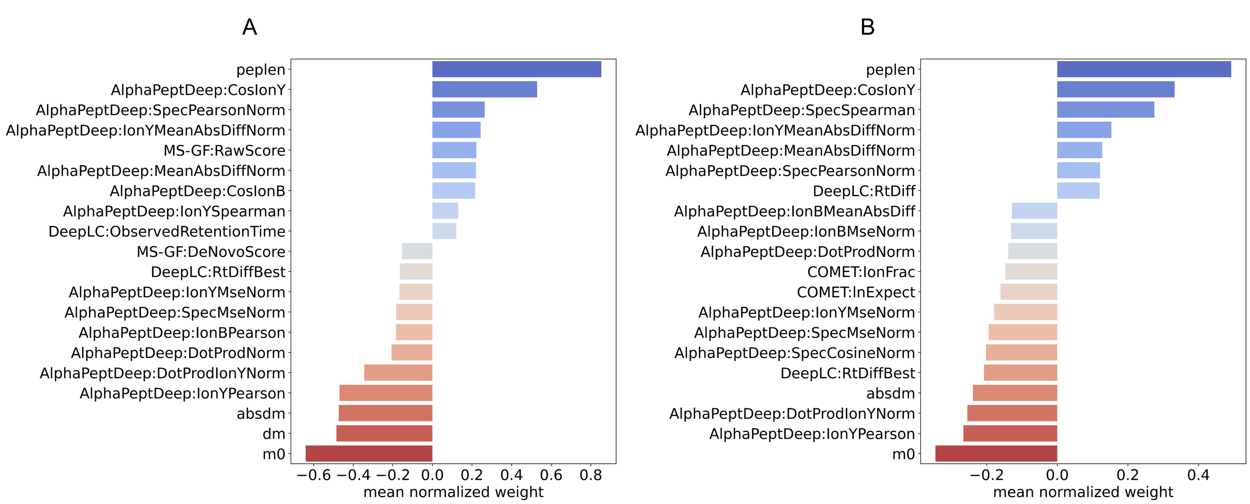

Figure S5: The top 20 normalized weights from Percolator for MSGF+ (A) and Comet (B) identification results in the PXD026824 dataset.

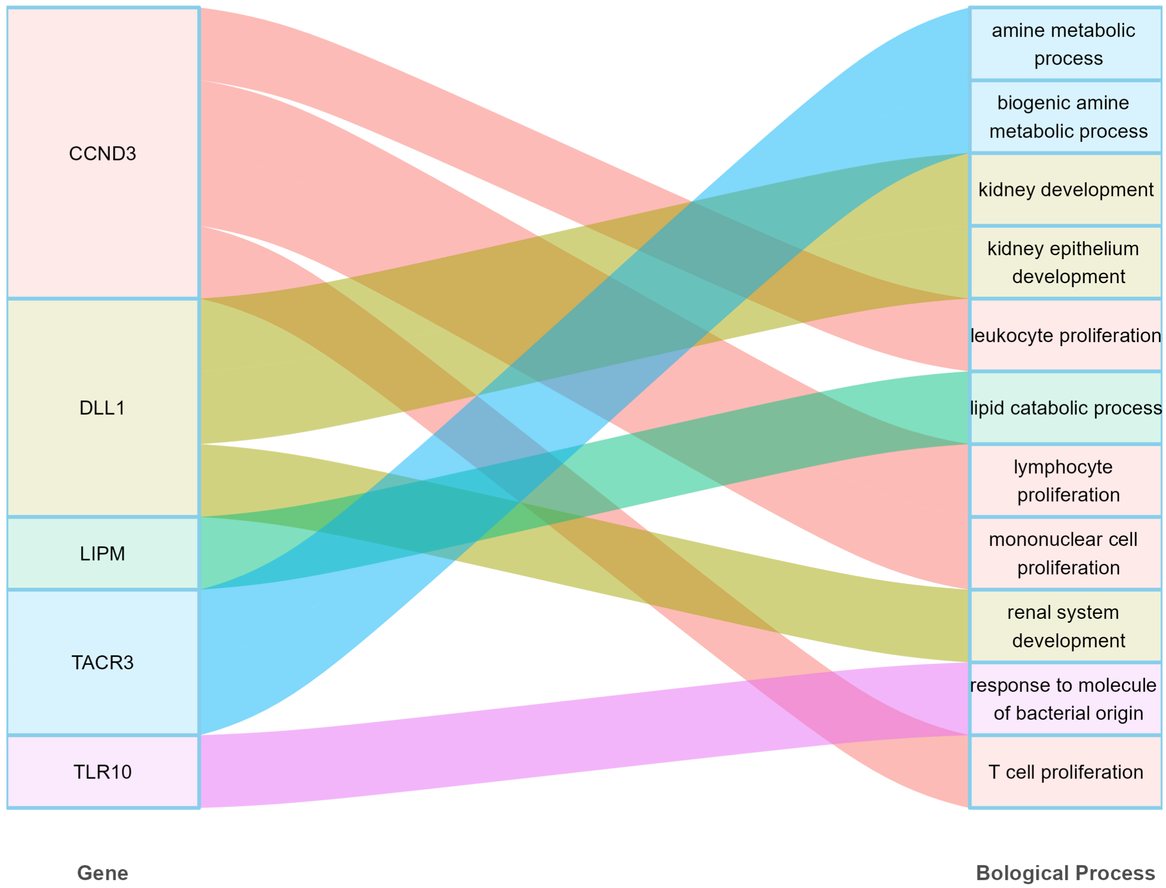

Figure S6: The GO enrichment analysis for MSGF+&Comet with quantms-rescoring. The Sankey diagram shows biological process term of the genes are uniquely reported by MSGF+&Comet with quantms-rescoring compared without quantms-rescoring.

Table S2: Runtime, memory, and CPU consumption for different configurations. The memory and CPU values represent the maximum resources required for a single job. The quantms workflow was run in a 64 cluster queue.

Datasets	Configuration	Runtime	Memory	CPU
PDC000127	Comet&MSGF+&MS²Rescore	9h39m	30G	7
PDC000127	Comet&MSGF+	6h28m	8G	4
PDC000127	Comet	5h10m	8G	4
PXD019643	Comet&MSGF+&MS²Rescore	15h57m	20G	8
PXD019643	Comet&MSGF+	13h09m	5.5G	4
PXD019643	Comet	2h05m	5.5G	4
PXD026824	Comet&MSGF+&MS²Rescore	2h05m	66G	8
PXD026824	Comet&MSGF+	1h17m	48.4G	4
PXD026824	Comet	24m	48.4G	4

Table S3: Experiment metadata and search tolerances (from SDRF). Columns: project accession, analytical method, instrument, fragmentation, tolerances (precursor/fragment), protease, modifications.

Project	Method	Instrument	Fragmentation	Tolerances	Protease	Modifications
PXD001819	LFQ	LTQ O. Velos	CID	20 ppm 0.5 Da	Trypsin/P	Carb (C) Ox (M) Ac
PDC000127	TMT	O. Fusion Lumos	HCD	20 ppm 20 ppm	Trypsin	Carb (C) TMT Ox (M) Ac
PXD019643	Immuno	O. Fusion Lumos	HCD	5 ppm 0.02 Da	Unspecific	Carb (C) Ox (M)
PXD026824	Phospho	O. Fusion Lumos	HCD	20 ppm 0.03 Da	Trypsin	Carb (C) Ox (M) Ac Phos

Acronyms: Carb = Carbamidomethyl; Ox = Oxidation; Ac = Acetyl; Phos = Phosphorylation; TMT = Tandem Mass Tag.